

Drift Index Definitions

KJV 1611 / KJV 1769 Semantic Drift Study

Purpose

This document defines the summary indices that convert unit level drift values into interpretable study results. These indices provide headline stability metrics, tail sensitive summaries, representation sensitivity summaries, and baseline separation metrics.

Guiding Principle

Embedding distance distributions in historical corpora are expected to be non-normal, right skewed, and tail sensitive. Primary indices emphasize robust quantities such as the median, median absolute deviation, and upper percentiles. Means and standard deviations are reported as secondary descriptors.

Primary Descriptive Indices

For each scope and representation, the study reports `n`, `median_drift`, `mad_drift`, `p95_drift`, `p99_drift`, `mean_drift`, `std_drift`, `min_drift`, and `max_drift`. Scopes include the shared canon overall, Old Testament shared corpus, New Testament shared corpus, and individual works.

Central Semantic Stability Index

The Central Semantic Stability Index (CSSI) is a centered transformation of cosine distance equivalent to the median cosine similarity between aligned text units under the cosine distance convention used in the study.

$$\text{CSSI} = 1 - \text{median_drift}$$

Tail Semantic Stability Indices

Tail Semantic Stability Indices measure stability at high drift percentiles. The study reports TSI at the 95th and 99th percentiles to characterize the highest drift regions of the corpus.

$$\text{TSI_P95} = 1 - \text{P95_drift}$$

$$\text{TSI_P99} = 1 - \text{P99_drift}$$

Representation Sensitivity Index

For matched units, representation sensitivity is measured as the difference between R0 and R1 drift. Large positive values indicate that the current normalization profile reduced apparent drift. Near zero values indicate that drift persisted after light normalization. Negative values are treated as inspection candidates.

Stability Gap Index

The Stability Gap Index (SGI) quantifies the separation between observed aligned drift and randomized baseline drift for a given scope, null model, and representation.

- $SGI = median_null - median_observed$
- $SGI_norm = SGI / median_null$

Study Decisions

Component	Study Decision
Primary headline metric	CSSI
Tail metrics	TSI_P95, TSI_P99
Sensitivity metric	Delta between R0 and R1 drift
Baseline separation metric	SGI, normalized SGI
Robust statistics	Median, MAD, P95, P99
Reporting hierarchy	Intratextual, intracorporeal, supracanonical

Expected Output Artifacts

- drift_indices_by_scope.csv
- sensitivity_indices_by_scope.csv
- baseline_gap_indices_by_scope.csv